Foundations of Computational Linguistics - Lecture 01

Lecture overview

Lecture 01 introduces Computational Linguistics (CL) as the scientific study of language production, comprehension, and acquisition, distinguishing it from Natural Language Processing (NLP), which focuses on engineering practical applications like machine translation and sentiment analysis. The course traces the field's evolution through three historical paradigms: the symbolic approach relying on handcrafted rules, the statistical approach utilizing machine learning with manual feature engineering, and the current neural approach that automatically learns patterns from raw data without human-defined features. Central to the lecture is the principle of compositionality, where meaning derives from individual signs and their specific combinations, alongside the challenge of resolving ambiguity through context, as illustrated by the "chicken" example distinguishing between verb and adjective forms.

The session further contextualizes CL within the broader history of linguistics, moving from ancient grammatical descriptions to the 19th-century discovery of Proto-Indo-European roots and early 20th-century structuralism. Key theoretical concepts include Saussure's arbitrariness of signs, the Sapir-Whorf hypothesis on linguistic relativity, and Chomsky's generative grammar, which differentiates between ideal linguistic competence and actual performance. The course objectives combine theoretical knowledge with practical Python programming skills to address modeling challenges such as segmentation, syntax, and semantics. Assessment is conducted entirely through a final project rather than an exam, with students expected to critically reflect on societal impacts, bias, and the limitations of current models while applying these concepts to self-chosen NLP tasks.

Slide 1: faculty of arts

This lecture introduces Computational Linguistics within the Faculty of Arts. It is the first session of the course, presented by Tim Van de Cruys on Tuesday, 22 September 2026. The slide serves as the title page establishing the course identity and initial meeting details.

Slide 2: Modeling language with computers

Modeling language with computers involves two perspectives: computational linguistics, the scientific study of linguistic production, comprehension, and acquisition; and natural language processing, the engineering application of computers to perform useful tasks like question answering, sentiment analysis, and machine translation. These fields represent two sides of the same coin, aiming to understand how language works while building practical applications that process or generate human language.

Slide 3: Three paradigms

Computational linguistics has evolved through three paradigms: the symbolic approach (1950s-1990s) uses handcrafted rules, dictionaries, and grammars; the statistical approach (late 1990s-2010s) relies on machine learning with large annotated corpora and manual feature engineering; and the neural approach (post-2010) utilizes neural networks on large raw or unannotated data without feature engineering. The key distinction is that while statistical methods require humans to define specific features, neural models automatically learn relevant patterns directly from raw text.

Slide 4: Applications

Sentiment analysis is a key application in computational linguistics that involves automatically identifying opinions regarding products, brands, or social issues. This process enables systems to detect and categorize subjective attitudes expressed within text data without human intervention.

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Slide 7: Applications

This slide highlights key computational linguistics applications, specifically sentiment analysis and machine translation. Sentiment analysis automatically predicts whether text, such as online star reviews, expresses positive, negative, or neutral opinions by assigning a score. Machine translation, exemplified by tools like Google Translate and DeepL, is another critical application that has been historically significant within the domain.

Slide 8: Applications

This slide outlines two primary applications of summarization in computational linguistics: the synthesis of multiple documents and simplification. Synthesis involves combining information from various sources into a single coherent output, while simplification focuses on reducing complexity to enhance readability. These concepts represent key functional areas where summarization techniques are applied within the field.

Slide 9: Applications

This page names ChatGPT as an application of large language models. An LLM generates text by predicting likely continuations from context, which supports many language tasks. Fluent output alone does not guarantee factual accuracy or establish human-like understanding.

Slide 10: The role of linguistics

This slide explores how computational models represent linguistic phenomena, specifically questioning how words are modeled, combined into sentences, and used to construct meaning representations. Using the example "The man eats an apple," it contrasts human understanding of word meanings with the mechanisms language models use to derive sentence-level meaning. Additionally, it investigates how existing linguistic theory can be harnessed to improve natural language processing applications.

Slide 11: Learning Objectives

This course aims to establish theoretical knowledge of computational linguistic analysis and natural language processing, including key research questions and applications. Practically, students will develop basic Python programming skills to apply existing algorithms and modules for specific research inquiries. Finally, the curriculum emphasizes critical reflection on the limitations and societal impacts of language technology, specifically addressing issues like bias and interpretability in model training.

Slide 12: Python

The course utilizes Python for hands-on programming exercises, where prior knowledge is beneficial but not mandatory. Students without experience will complete a self-study interactive notebook titled "Introduction to Python for Linguists" to master basic concepts before practical sessions. These notebooks provide an interactive environment combining executable code with explanatory text, allowing students to experiment with and observe the results of various language and NLP models.

Slide 13: Notebook

A notebook is an interactive environment combining text, code, and results, with two primary options: local installation via Jupyter Notebook/Lab or the web-based Google Colab interface requiring a Google account. While local setup allows running models on your own computer, Google Colab executes code on remote servers provided by Google. The professor recommends attempting the local installation for a more engaging experience, though the web interface serves as an alternative if local setup is too difficult.

Slide 14: Python: installation

While the standard Python version from python.org is available, it can be cumbersome to install, particularly on Windows. The professor recommends using the Anaconda distribution (anaconda.com) instead, as it is a user-friendly environment pre-installed with modules suitable for data science and machine learning. Anaconda simplifies package management and facilitates the easy installation of Jupyter Notebook or Jupyter Lab, though the specific installer depends on your operating system.

Slide 15: Practical Information

Full course materials, including slides and additional reading, are accessible online via Toledo. Lectures are recorded and made available on the platform, provided that infrastructure permits recording. For further inquiries regarding Computational Linguistics, contact tim.vandecruys@kuleuven.be.

Slide 16: Evaluation

This course features no exam; evaluation relies entirely on a final project where students apply acquired theoretical and practical knowledge to a self-chosen NLP application or computational linguistic research question. The professor emphasizes that this project integrates both model theory and practical usage, with further details provided in upcoming lectures. Additionally, the course is offered in two versions: a 4 ECTS option and a 6 ECTS option.

Slide 17: Topics

This course covers machine learning for language, neural networks, language modeling, grammar models, semantics, attention mechanisms, and transformers. It also addresses machine translation, multimodality, interpretability, bias, and the role of linguistics. While AI students typically follow a 4 ECTS track, Digital Humanities and Linguistics students may take a 6 ECTS version that includes an additional lecture on applications in these fields.

Slide 18: Language use: an example

Using the joke "Why did the chicken cross the road?", this slide illustrates how language use requires interpreting words within context, such as distinguishing the verb "cross" (to go across) from the adjective meaning angry. It further demonstrates that generating correct sentences depends on applying grammatical rules for verb forms, specifically using the infinitive "cross" rather than the past tense "crossed," and adhering to standard word order instead of incorrect structures like "why the chicken did cross."

Slide 19: Language use: an example

The complete joke asks why the chicken crossed the road and gives the literal answer, “to get to the other side.” It illustrates how a listener uses the ordinary expectation that a why-question asks for a motive, while the punchline supplies a deliberately literal explanation.

Slide 20: Language use: an example

This animation build corrupts the opening words of the chicken question (“Wide id…”). The rest of the exchange remains recognizable, showing that readers can recover intended language despite a noisy or ill-formed string.

Slide 21: Language use: an example

The bold focus moves to “chicken cross.” The sentence meaning depends on identifying the subject and the event, not on treating every word independently.

Slide 22: Language use: an example

The bold focus is “cross,” the action in the question. The image and the punchline make the literal route across the road salient.

Slide 23: Language use: an example

“Why did the chicken crossed…” adds an ungrammatical past-tense ending after the auxiliary did. The intended question remains recoverable, but grammatical form affects acceptability.

Slide 24: Language use: an example

“Why the chicken did cross…” changes the usual English question word order. It remains interpretable but is marked; the slide contrasts a well-formed question with a rearranged version.

Slide 25: Language use: an example

The bold reply “I give up” makes the listener’s expected search for a motive explicit. That expectation sets up the literal punchline on the following build.

Slide 26: Language use: an example

The bold punchline is “To get to the other side.” The joke works because the expected explanation is replaced by a literal description of the route.

Slide 27: What is language?

A language is a communication system that pairs forms with meanings and lets people use them in context. The following sign examples distinguish resemblance-based, trace-based, and convention-based form-meaning relations.

Slide 28: What is language?

Language is defined as a communication system where the key element is a sign, which pairs a form with a meaning. These signs can be iconic, indexical, or symbolic. Understanding language requires significant knowledge, as illustrated by a joke example where interpreting the exchange depends on recognizing the context of spoken conversation rather than just the literal words.

Slide 29: Slide 29

The bicycle drawing is an iconic sign: its shape resembles the object it refers to. This example introduces the relation between a sign's form and its meaning.

Slide 30: Slide 30

Footprints are an indexical sign because they are a trace causally connected to the animal that made them. The viewer can infer a prior presence from the mark.

Slide 31: Slide 31

The radiation trefoil is a symbolic sign. Its shape does not resemble radiation; people understand it through a learned social convention.

Slide 32: apple

The written word apple is a symbolic sign. The letters do not resemble the fruit; speakers learn the conventional link between this word form and its meaning.

Slide 33: penguin

Penguin illustrates the same arbitrary form-to-meaning relation. The spelling does not resemble the animal, so readers rely on a language convention they have learned.

Slide 34: hypothesis

Hypothesis is another symbolic sign. The word's form does not reveal the abstract concept; its meaning comes from the shared vocabulary of the language.

Slide 35: What is language?

Language is defined as a communication system where the key element is the sign. A sign consists of the pairing of a form and a meaning, which can be iconic, indexical, or symbolic. This framework establishes the fundamental structure of language within computational linguistics.

Slide 36: What is language?

Language is defined as a communication system based on signs, where a sign represents the pairing of a form and a meaning. These signs can be iconic, indexical, or symbolic, with the specific meaning often determined by context and global language structure rather than a direct link between form and meaning.

Slide 37: Animal vs. human communication

Slide 37 outlines key features of human language, including interchangeability, cultural transmission, arbitrariness, discreteness, displacement, and productivity. While some animal systems share these traits, others are considered exclusively human. Interchangeability refers to bi-directional communication where speakers can also hear; cultural transmission indicates language is learned during childhood rather than innate; and arbitrariness means linguistic signs lack an inherent connection to their meanings.

Slide 38: Vervet monkey

Vervet monkeys use arbitrary signs to signal specific predators, functioning as true communication rather than mere expressions of stress or excitement. These signs possess a stable external reference and include three distinct vocalizations: a short two-syllable coughing sound, a barking or peeping sound, and a continuous clicking 'chutter' sound, each corresponding to a different type of predator.

Slide 39: Vervet monkey

Vervet monkeys use arbitrary signs with stable external references to communicate specific predator threats, rather than merely expressing stress. They employ three distinct calls: a short coughing sound for eagles (prompting looking up and seeking shelter), a barking or peeping sound for leopards (triggering escape into trees), and a continuous clicking 'chutter' for snakes (causing them to stand up and look down). These symbolic signals function like words by conveying detailed information that dictates specific escape plans.

Slide 40: Human language: discreteness

Discreteness, also known as duality of patterning or double articulation, describes how human language constructs meaningful signs from discrete, meaningless units like sounds. For example, the sounds in "stop" can be reorganized to form new words such as "spot," "tops," or "opts." This two-level structure contrasts with animal communication systems, which primarily rely on indivisible sign units that cannot be broken down and recombined.

Slide 41: Human language: displacement

Displacement is a defining feature of human language that allows speakers to discuss entities or events not present in the immediate space or time. Examples include referencing past occurrences like "last night," distant locations such as "a man on the moon," fictional concepts like "a unicorn," or future dates like "September 2037." In contrast, animal communication systems are strictly restricted to the present moment and the immediate environment.

Slide 42: Human language: productivity

Human language exhibits productivity, enabling humans to produce and understand an infinite number of novel utterances through creative combinations. This process builds hierarchically from phonemes to morphemes, words, phrases, and sentences, allowing for new terms like "Brexit" or extended descriptions. In contrast, animal communication is limited to a single sign per utterance, lacking this generative capacity.

Slide 43: Compositionality

Frege's principle of compositionality states that an utterance's meaning derives from individual signs and their specific combination, allowing us to understand novel sentences like "Jim drives the car" versus "The car drives Jim." This is complemented by co-compositionality, where combined signs mutually influence each other's meaning. For instance, the word "red" shifts its specific shade or connotation depending on whether it modifies a rose, wine, or hair.

Slide 44: How to make sense of human speech?

Making sense of human speech requires segmenting utterances into basic sound units (phonemes) and meaningful units (morphemes), then composing them into larger structures via syntax. Next, compositional semantics derives meaning from individual words and their combinations, while contextualization ensures full interpretation by accounting for the specific conversational situation. This process transforms isolated sentences into pragmatically meaningful communication, as demonstrated by how context alters the understanding of jokes or complex terms like "TikTokification."

Slide 45: Linguistics

Linguistics is the study of language, examining all aspects of human language use and the knowledge required to employ it. It addresses three core questions: how languages are structured, how they are used (sociolinguistics), and how they change over time (diachronic linguistics). While this course primarily focuses on modeling language structure computationally, it generally analyzes language in a synchronic context rather than its historical evolution.

Slide 46: Linguistics

Linguistics is the study of language, focusing on all aspects of human language use and the underlying knowledge required to employ it. The discipline investigates three primary questions: how languages are structured, how they are used in practice, and how they change over time.

Slide 47: A brief history of linguistics

Linguistics originated in ancient India with Pāṇini's Sanskrit grammar descriptions, which remain relevant today. In Greece and Rome, grammars for Greek and Latin were developed primarily for rhetorical purposes to ensure effective language use. During the Middle Ages, language study formed part of the trivium (grammar, logic, rhetoric) within the seven liberal arts, while the Renaissance shifted focus toward standardizing vernacular languages beyond Greek and Latin.

Slide 48: Birth of linguistics as a science

Linguistics emerged as a scientific field through diachronic research comparing historical language forms to identify common roots. In 1786, William Jones recognized shared origins among Sanskrit, Greek, Latin, and other European languages, leading to 19th-century historical and comparative linguistics by figures like the Brothers Grimm. This approach enabled the reconstruction of Proto-Indo-European and its descendant language families, such as Germanic, Romance, Slavic, and Celtic, based on systematic similarities across variants.

Slide 49: The Indo-European Language Family

The Indo-European language family illustrates how diverse modern languages evolved from a single common ancestor, Proto-Indo-European, through a branching tree structure. Linguists reconstruct this history by comparing cognates, such as the word for "father," which appears with an F/V sound in Germanic languages (e.g., English) but a P sound in Romance languages (e.g., Latin pater). Although the exact pronunciation of Proto-Indo-European remains unknown due to a lack of written attestation, its existence is inferred from these systematic relationships across Western European and other Indo-European branches.

Slide 50: First half of the 20th century

In the first half of the 20th century, European Structuralism (Saussure) posited that sign meaning is determined by relationships within a language structure, emphasizing the arbitrariness of signs. Concurrently, American Descriptive Linguistics (Sapir, Whorf, Bloomfield) studied diverse Native American languages to highlight structural differences from European tongues. This research led to linguistic relativity, or the Sapir-Whorf thesis, which asserts that grammatical systems and language use directly influence human thought and perception of reality.

Slide 51: Noam Chomsky

Noam Chomsky introduced generative grammar, a formal system of rules generating all grammatical sentences for a language, which significantly influenced computational linguistics. He distinguished between ideal linguistic knowledge (competence) and actual usage (performance), arguing that innate linguistic universals exist within a Language Acquisition Device. This innate system explains the discrepancy between the complexity of natural languages and the ease with which children acquire them from sparse input.

Slide 52: Relevance for our course

This course examines computational linguistics through theoretical, practical, and critical lenses, acknowledging its roots in Chomskyan formalization and recent structuralist revisits. It addresses why language is difficult for computers due to its complex interaction of knowledge types and the ubiquitous challenge of ambiguity. Additionally, the curriculum emphasizes understanding model limitations, specifically regarding bias and the lack of interpretability in modern systems.

Slide 53: Levels of linguistic structure

Linguistic structure comprises five levels: phonology (basic meaningless sound units), morphology (word construction using suffixes like in "TikTokification"), syntax (organizing words into sentences), semantics (combining word meanings into larger sentence meanings), and pragmatics (interpreting utterances within a discourse context). These hierarchical levels are essential for modeling language and resolving the inherent ambiguities found at each stage.

Slide 54: Ambiguity: phonology

Phonological ambiguity arises from homophones, which are distinct words that share the same pronunciation, such as "buy," "by," and "bye." This type of ambiguity is a critical challenge in speech recognition systems, where acoustic input must be correctly mapped to specific written forms despite identical sounds.

Slide 55: Ambiguity: morphology

Morphological ambiguity arises from homonyms or homographs where a single word form represents different parts of speech, such as "flies" (noun/verb), "past" (preposition/noun), or "like" (preposition/verb). This phenomenon complicates both syntactic and semantic analysis, particularly when multiple ambiguities combine within a sentence. The classic examples "Time flies like an arrow" and "Fruit flies like a banana" demonstrate how identical word sequences can yield entirely different meanings depending on the grammatical interpretation of each term.

Slide 56: Ambiguity: syntax

This slide illustrates structural ambiguity in syntax using two sentences: "John hits the ball with a bat" and "John hits the man with a hat." Both examples share an identical dependency structure consisting of a subject noun phrase (NP-SBJ), a dependent verb (DEP), an object noun phrase (NP-OBJ), and a prepositional phrase (PP) containing another noun phrase. The slide highlights how this specific syntactic arrangement can lead to different interpretations despite the same surface-level grammatical components.

Slide 57: Ambiguity: semantics

This slide illustrates semantic ambiguity through the word "pen," which carries distinct meanings depending on context. The examples demonstrate how "pen" can refer to a writing instrument, an enclosure for animals, a prison, or the action of writing. These variations highlight how identical lexical forms can possess multiple unrelated definitions within different sentences.

Slide 58: Ambiguity: pragmatics

Pragmatic ambiguity arises when context is required to resolve pronoun references or causal relationships, as illustrated by examples where "His" refers to Joe's office and "The first lady" implies Jill. Additionally, the sentence regarding monkeys not eating bananas demonstrates how the antecedent of "they" shifts meaning based on whether the cause is hunger or ripeness. These cases highlight that semantic structure alone is insufficient without pragmatic inference to determine the correct interpretation.

Slide 59: Difficulties

Machine translation faces significant difficulties due to differing word orders between languages, specifically the modifier-noun structure in English versus the noun-modifier structure in French. This syntactic divergence is illustrated by translating "a red house" to "une maison rouge" and the more complex phrase "almost double the usual oatmeal biscuits" to "presque le double des biscuits à l'avoine habituels."

Slide 60: Difficulties

Machine translation faces significant difficulties because words often possess multiple meanings, creating ambiguity for automated systems. A primary example is the French word "avocat," which can translate to either "lawyer" or "avocado" depending on context. This polysemy illustrates the core challenge of accurately determining the correct meaning without additional contextual information.

Slide 61: Difficulties

Machine translation faces significant difficulties because words often possess multiple distinct meanings. A primary example is the French word "avocat," which illustrates this ambiguity as it can translate to either "lawyer" or "avocado" depending on context. This polysemy presents a core challenge for computational systems attempting to determine the correct interpretation without additional contextual cues.

Slide 62: Difficulties

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Slide 63: Difficulties

Machine translation faces significant difficulties because a single word can possess multiple distinct meanings depending on context. This ambiguity is illustrated by the French word "avocat," which translates to either "avocado" or "lawyer" in English, demonstrating the challenge of selecting the correct semantic interpretation without additional contextual cues.

Slide 64: Difficulties

Machine translation faces significant difficulties due to lexical ambiguity, where a single source word maps to multiple target meanings depending on context. The slide illustrates this with French terms like "pied" (foot/paw) and "étape" (journey/step), which require disambiguation based on whether the subject is human or animal. Similarly, words such as "leg," "jambe," and "patte" demonstrate how semantic distinctions between humans and animals complicate accurate translation.

Slide 65: Difficulties

Machine translation faces difficulties in languages like Spanish where subject pronouns are often omitted. Instead of explicit subjects, verb conjugations indicate the person through specific endings: -o (I), -as (you), -a (he/she/it), -amos (we), and -an (they). This creates ambiguity for systems that must determine whether to translate the third-person singular ending as "he," "she," or "it" without an explicit subject pronoun.

Slide 66: Difficulties

Machine translation faces significant difficulties with pronoun resolution, as demonstrated by the ambiguous sentence "The monkeys did not eat the bananas because they were not [adjective]." In French, the correct gender agreement for the pronoun "they" changes based on whether the antecedent is the monkeys ("ils") or the bananas ("elles"), illustrating how context determines meaning. This example highlights the challenge of disambiguating references without explicit semantic understanding.

Slide 67: Difficulties

This slide illustrates machine translation difficulties through English sentences containing the pronoun "her" and their German translations, which vary based on grammatical case. The examples show that "her" translates to "sie" (accusative) in "I saw her," but changes to "ihr" (dative) in "I gave her a present" and "ihrem" (dative genitive) when modifying "brother." These variations highlight the challenge of correctly mapping pronouns across languages with different case systems.

Slide 68: Difficulties

The French present-tense sentence “je bois du lait” is the source for two possible English renderings on the next builds. Its tense/aspect does not force a single English choice without context.

Slide 69: Difficulties

“I drink milk” is one translation of “je bois du lait.” It can express a habitual or general reading; context may instead call for the progressive form shown on the next page.

Slide 70: Difficulties

“I’m drinking milk” is another possible translation of the French present. The pair demonstrates that tense and aspect do not map one-to-one across languages, so surrounding context can decide between them.

Slide 71: Difficulties

The examples “bone of contention,” “kick the bucket,” and the bear-fur proverb have conventional meanings that differ from literal word-by-word interpretation. Translation must preserve the intended expression, often using a target-language idiom or paraphrase.

Slide 72: Difficulties

The spell-checking examples contrast too and to. Both are valid words, so correcting the sentence requires syntax and meaning: “too hot to drink” uses too as an intensifier and to as an infinitive marker.

Slide 73: Difficulties

Question answering maps a natural-language question to an answer supported by relevant information. This slide marks that application area; it does not yet show a specific question or answer.

Slide 74: Difficulties

This second build remains the Question Answering section marker. The task requires identifying what information the question requests and locating or deriving a supported answer; no worked example appears on this page.

Slide 75: Difficulties

Speech technology must handle ambiguity in both recognition and synthesis. A recognizer may confuse “recognize speech” with the similar-sounding phrase “wreck a nice peach”; a synthesizer must use context to pronounce words such as content and address with the right stress.

Slide 76: Difficulties

This build adds spellings such as “rough,” “women,” and “nation” to show that English spelling does not map transparently to sound. Recognition uses acoustic context to distinguish phrases, while synthesis needs lexical and syntactic context to choose pronunciation.

Slide 77: Difficulties

The final build restores the intended phrase “recognize speech.” It contrasts speech recognition, which maps sound to words, with speech synthesis, which must select pronunciations such as the noun or verb stress in content and address.

Key points to remember

The session further contextualizes CL within the broader history of linguistics, moving from ancient grammatical descriptions to the 19th-century discovery of Proto-Indo-European roots and early 20th-century structuralism. Key theoretical concepts include Saussure's arbitrariness of signs, the Sapir-Whorf hypothesis on linguistic relativity, and Chomsky's generative grammar, which differentiates between ideal linguistic competence and actual performance. The course objectives combine theoretical knowledge with practical Python programming skills to address modeling challenges such as segmentation, syntax, and semantics. Assessment is conducted entirely through a final project rather than an exam, with students expected to critically reflect on societal impacts, bias, and the limitations of current models while applying these concepts to self-chosen NLP tasks.